Ashrai Gandikota

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Objective

Computer Engineering student specializing in embedded systems and scalable software development,hardware-level programming with modern web development experience. Versatile at navigating both low-level embedded hardware constraints and scalable full-stack cloud architectures, engineering optimized solutions ranging from C-based memory allocation to responsive web components. Active contributor to full-stack web applications and geospatial tools serving the university student community. Seeking a software development internship for Fall 2026.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
Bachelor of Science in Computer Engineering, GPA 4.00 Expected Graduation, May 2028

Skills

Programming: Java, Python, C, C#, Assembly(RISC-V), JavaScript, SQL, HTML, MATLAB
Platforms: Linux (Ubuntu, Kali, Debian)
Hardware: Raspberry Pi, ARM Mbed microcontroller, ARM Cortex-M3 (LPC1768), Oscilloscope
Software: Github, AWS, MySQL, Figma, Docker, VMware
Languages: Telugu (native), English (fluent), Spanish (beginner)

Experience

Baazar | Remote *June – August 2025*
Position / Department (if applicable)
A cloud based e-commerce platform that uses modern web technologies and AWS infrastructure to deliver highly scalable, high-performance digital retail.

- Assisted in migration and optimization of the e-commerce platform, reducing infrastructure costs by ~15% and improving page load speeds by~27%.
- Built integrations between the Shopify API and AWS Aurora (mySQL) to support scalable product, order, and customer workflows.
- Developed responsive components in React, collaborating through Figma to translate UI/UX designs into production-ready features.

Projects

Buzz Blitz! | Electrical and Computer Engineering, Project 2, Programming Hardware/Software Systems *Spring 2026*
Sole Contributor
ESP32 microcontroller project that showcased skills in complex data structures, managing memory, and interfacing hardware with software in C. The objective was to create a 2D side-scroller, similar to games like jetpack joyride, with advanced features.

- Programmed core gameplay mechanics in C, including dynamic obstacle generation (spiderwebs), weapon switching, and projectile physics (piercing stingers).
- Engineered a robust memory allocation strategy to support continuous random object spawning and custom replay functionality, successfully resolving memory leak-induced crashes and watchdog timer resets.
- Interfaced physical hardware pushbuttons for real-time character movement and implemented complex game-state controls, including a timed pause menu and a high-score leaderboard system.

Relevant Coursework

Programming Hardware/Software Systems: Low-level embedded programming; RISC-V assembly and dynamic programming algorithms; C memory management and stack/heap allocation logic
Introduction to Signal Processing: MATLAB signal processing; frequency-domain transformations; phasor-domain signal analysis
Circuit Analysis: Mesh currents and node voltages; Laplace transforms; hardware configuration and equivalent resistance

Leadership or Activities

GT Web Development | Buzz Board Developer *August 2025 – May 2026*

- Co-developed "Buzz Board," a full-stack campus event platform allowing students to discover, track, and navigate to Georgia Tech events in real-time.